**FYUGP B.Sc. PHYSICS HONOURS**

ELECTRODYNAMICS II

Semester V · Core in Major

**MODULE 2: ELECTROMAGNETIC INDUCTION AND MAXWELL'S EQUATIONS**

Model questions on topics the past papers miss

## Section A: 1 mark each, answer all

***Motional emf and emf***

1 The emf induced across a conductor of length $l$ moving with speed $v$ perpendicular to a uniform field $B$ is _____. **(1 mark)**

***Induced electric field***

2 State whether true or false: the electric field produced by a changing magnetic flux is conservative. **(1 mark)**

3 Write the Maxwell–Faraday law in differential form. **(1 mark)**

***Ohm's law and displacement current***

4 Write Ohm's law in differential form. **(1 mark)**

5 Write the displacement current density in terms of $\vec{E}$. **(1 mark)**

***Energy in magnetic fields***

6 Write the magnetic energy density in a linear medium in terms of $\vec{H}$ and $\vec{B}$. **(1 mark)**

## Section B: 2 marks each, answer all

***Motional emf***

7 Why does a conductor moving across a magnetic field develop an emf when no circuit flux changes? Give the force responsible. **(2 marks)**

***Induced electric field***

8 Distinguish the electric field of static charges from the induced electric field produced by a changing magnetic field. **(2 marks)**

***Faraday's law***

9 State the physical meaning of the minus sign in Faraday's law. **(2 marks)**

***Displacement current***

10 A capacitor is charged by a steady current. Explain why Ampère's law as written fails for a surface passing through the gap, and how Maxwell fixed it. **(2 marks)**

11 Write the integral form of Ampère–Maxwell law and identify the term Maxwell added. **(2 marks)**

## Section C: 4 marks each, answer any two

***Motional emf and Ohm's law***

12 A conducting bar of length 0.5 m slides at 2 m/s on frictionless parallel rails in a uniform field of 0.4 T. The rails are joined by a resistance of 0.2 $\Omega$. Find the emf, the current, the force needed to keep the bar at constant speed, and the power dissipated. **(4 marks)**

***Displacement current***

13 A parallel-plate capacitor has plates of area 0.02 $m^2$ separated by 2 mm. The voltage across it rises at $5 \times 10^4$ V/s. Find the displacement current between the plates, and the conduction current in the wire. Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m. **(4 marks)**

***Energy in magnetic fields***

14 A long solenoid has 500 turns per metre, radius 2 cm and carries a current of 2 A. Find the magnetic energy stored in a length of 0.5 m. Take $\mu_0 = 4\pi \times 10^{-7}$ H/m. **(4 marks)**

***Induced electric field***

15 A long solenoid of radius 5 cm has $dB/dt = 0.1$ T/s inside it, and zero field outside. Find the magnitude of the induced electric field at $r = 2$ cm and at $r = 8$ cm from the axis. **(4 marks)**

## Section D: 10 marks each, answer any one

***Maxwell's equations in matter***

16 Starting from the Maxwell equations, derive $\nabla \cdot \vec{D} = \rho_f$ and $\nabla \times \vec{H} = \vec{J}_f + \partial \vec{D}/\partial t$. Define $\vec{D}$ and $\vec{H}$ in terms of $\vec{E}$, $\vec{P}$, $\vec{B}$ and $\vec{M}$. State the boundary conditions for the normal component of $\vec{D}$ and the tangential component of $\vec{H}$ at a surface carrying free current. **(10 marks)**

***Faraday's law***

17 Explain the difference between the transformer emf and the motional emf. Show that $\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{a}$ holds for a loop that is moving as well as for one that is fixed. **(10 marks)**

**MODULE 2: MODEL ANSWERS**

For checking your work. Each answer shows the key step and the result.

**Section A.** A1 $vBl$. A2 False: its line integral round a closed loop is not zero. A3 $\nabla \times \vec{E} = -\partial \vec{B}/\partial t$. A4 $\vec{J} = \sigma \vec{E}$. A5 $\vec{J}_d = \varepsilon_0 \, \partial \vec{E}/\partial t$. A6 $u = \frac{1}{2} \vec{H} \cdot \vec{B}$.

**Section B.** B1 The magnetic force $q\vec{v} \times \vec{B}$ pushes positive charges along the bar, so an emf $\oint (\vec{v} \times \vec{B}) \cdot d\vec{l}$ appears. B2 Static-charge fields are conservative ($\nabla \times \vec{E} = 0$); the induced field has $\nabla \times \vec{E} = -\partial \vec{B}/\partial t \neq 0$. B3 Through the gap $I = 0$, so $\mu_0 I$ alone gives zero; Maxwell added $\mu_0 \varepsilon_0 \, d\Phi_E/dt$, which fixes it. B4 Lenz: the induced emf opposes the change in flux. B5 $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} + \mu_0 \varepsilon_0 \frac{d}{dt} \int \vec{E} \cdot d\vec{a}$.

**Section C.** C1 emf $= Blv = 0.4 \times 0.5 \times 2 = 0.4$ V; $I = 0.4/0.2 = 2$ A; $F = BIl = 0.4 \times 2 \times 0.5 = 0.4$ N; power $= Fv = 0.8$ W $= I^2 R$. C2 $C = \varepsilon_0 A/d = 8.85 \times 10^{-11}$ F; conduction current $I = C \, dV/dt = 4.43 \, \mu$A; displacement current between plates is the same, $4.43 \, \mu$A. C3 $B = \mu_0 n I = 1.257 \times 10^{-3}$ T; $u = B^2/2\mu_0 = 0.628$ J/m$^3$; volume $= \pi r^2 l = 6.28 \times 10^{-4}$ m$^3$; $U = 3.95 \times 10^{-4}$ J (check: $\frac{1}{2} L I^2$ with $L = 1.97 \times 10^{-4}$ H). C4 Inside, $E = \frac{r}{2} \, dB/dt = 1.0$ mV/m at $r = 2$ cm; outside, $E = \dfrac{R^2}{2r} \, dB/dt = 1.56$ mV/m at $r = 8$ cm.

**Section D.** D1 Gauss in $\vec{E}$ with $\rho = \rho_b + \rho_f$ and $\rho_b = -\nabla \cdot \vec{P}$ gives $\nabla \cdot \vec{D} = \rho_f$, with $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$. Ampère–Maxwell with $\vec{J} = \vec{J}_f + \nabla \times \vec{M} + \partial \vec{P}/\partial t$ gives $\nabla \times \vec{H} = \vec{J}_f + \partial \vec{D}/\partial t$, with $\vec{H} = \vec{B}/\mu_0 - \vec{M}$. Boundary conditions: $D^{\perp}_{above} - D^{\perp}_{below} = \sigma_f$ and $\vec{H}^{\parallel}_{above} - \vec{H}^{\parallel}_{below} = \vec{K}_f \times \hat{n}$. D2 Transformer emf comes from $\partial \vec{B}/\partial t$ with the loop fixed; motional emf from $\vec{v} \times \vec{B}$ with the loop moving. Both follow from $\oint \vec{E}_{\text{loop}} \cdot d\vec{l}$ in the frame of the loop, and the flux rule $\mathcal{E} = -d\Phi/dt$ covers both.

### Mapping to the syllabus and textbook

| Syllabus item | Topic | Griffiths section | Model questions |
|---|---|---|---|
| 6 | Motional emf | 7.1.3 | 1, 7, 12 |
| 6 | Emf of a source; Ohm's law $\vec{J} = \sigma \vec{E}$ | 7.1.1–7.1.2 | 4 |
| 7 | Induced electric field, not conservative | 7.2.2 | 2, 3, 8, 15 |
| 8 | Displacement current, numerical | 7.3.2 | 5, 10, 11, 13 |
| 7 | Energy in magnetic fields, $u = \frac{1}{2} \vec{H} \cdot \vec{B}$ | 7.2.4 | 6, 14 |
| 8 | Maxwell's equations in matter, $\vec{D}$ and $\vec{H}$ form | 7.3.5 | 16 |
| 7 | Transformer vs motional emf, flux rule | 7.2.1, 7.1.3 | 17 |